This is fictional examination staging, not a historical quotation or a mathematical source. The quiz, learner guide, instructor key, and manuscript govern the finite definitions. The bracket-display restriction below is a chalkboard convention; the paper uses JSON teaching packets.

# 00: The Staging, the Professor, and the Tribunal of Paper 000

**AuthorityEffect:** None.
**Location:** Moscow, Sadovaya 302-bis, Apartment 50.
**Presiding Tribunal:** Messire Woland, Lev Davidovich Landau ("Dau"), Koroviev-Fagotto, Behemoth, Azazello, Abadonna, Hella; with the Russian-American Delegation: Jeremy, Liudmila, Polina, Yaroslava.
**Subject of Examination:** Paper 000 — *The Symbol Game: A First Game with $S(n)$, MSS-N, and Basic Packets.*
**Companion Documents:** - Study Guide: study_guide.md - Examination Paper: quiz.md - Instructor Key: instructor_key.md

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## 1. The Knock at the Door

The cold autumn sleet over Sadovaya Street had turned into driving grey needles. Water gurgled in the rusted drainpipes of Apartment 50. Inside, the heavy crimson velvet drapes were bolted shut against the Moscow night. A fire roared in the tiled stove, smelling of birch bark, burnt kerosene, and sulfur.

You stood in the dimly lit hallway, soaked through your overcoat, clutching a leather folder filled with notes on smooth manifolds, continuous Lie groups, and elaborate multi-page JSON-RPC envelopes you had downloaded from contemporary AI preprints. In your notebook, you had carelessly scribbled:

"MSS-2 is obviously identical to $S(2)$"

You knocked twice with trembling knuckles.

The heavy oak door swung open without a sound.

Standing on the threshold was **Hella**. Her pale green eyes were cold and unblinking. Across her slender throat ran the thin, vivid crimson scar of the resurrected. She looked down at your hands, saw the leather folder stuffed with JSON envelopes and coordinate charts, and curled her upper lip with disgust:

> *"Leave that paperwork in the umbrella stand. For this chalkboard round, use the bracket display. Keep your JSON packets beside the*

*paper; they remain the companion format. If you mistake an envelope for its contents, I will confiscate your chalk."*

She stepped aside and pointed a long, bloodless finger toward the candlelit drawing room.

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## 2. Woland Introduces Dear Professor Landau

In the center of the drawing room sat **Messire Woland**, sunk into a high-backed walnut armchair. His right leg was stretched out onto an embroidered velvet footstool, wrapped in thick wool bandages at the knee. Resting between his knees was his heavy black walking cane, capped with the cast-silver head of a snarling poodle. One eye was black and bottomless; the other sparkled with a piercing, mocking green fire.

*"Look at him, Dau,"* Woland said, his deep bass voice vibrating through the crystal drops of the chandelier. *"Another ambitious traveler from the West. He wishes to play the Symbol Game. He believes that repeating words like 'quantum consciousness', 'differentiable manifold', and 'generative transformer' constitutes a license to create physical law."*

Woland tapped his silver cane against the floor.

*"Step out, Lev Davidovich. Let us see if our guest can distinguish two marks on a card without losing his mind."*

From the dark alcove near the grandfather clock stepped **Lev Davidovich Landau**—"Dau." Slight, narrow-shouldered, with rebellious black hair and intense, fiery dark eyes, Landau wore an unironed tweed jacket with chalk-dusted lapels. In his fingers, he spun a crisp cylindrical stick of railway chalk with the speed of an electric drill.

Landau stopped the chalk dead in his palm, leveled it at your face, and let out a sharp, dry snort:

> *"Well? What are you staring at? Have you come to take the Theoretical Minimum, or have you come to recite prose from a marketing brochure?
>
> Let me tell you plainly: I do not care about your university degrees, I do not care about your multi-billion-parameter language models, and I do not care about your six-layer JSON wrappers. In my seminar in Kharkiv, if a student confused a coordinate representation with an invariant tensor, I threw him out of the room. Here, Paper 000 asks the simplest question imaginable: you have two marks, $a$ and $b$. Can you state an irreflexive relation without confusing the proposal with the referee's receipt?

If you submit a self-pair under this board's irreflexive relation rule, Behemoth will set fire to your coat. Equality itself may remain reflexive. If you assert that MSS-2 is $S(2)$, Koroviev will laugh until dawn. And if you replace an exact capstone calculation with an unchecked approximation, Azazello will send you back to your arithmetic."*

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## 3. The Retinue and the Delegation

Around the salon, the rest of the tribunal stood at their assigned posts:

- **Koroviev (Fagot)** was balanced on a rickety three-legged stool, wearing dirty checkered trousers that ended six inches above his boots, an unbuttoned vest, and a cracked pince-nez attached to his ear by dirty thread. He giggled shrilly: > *"Oh, Messire! Marvel at this magnificent mind! Look at his pockets—crammed full of 'intuitive prompts'! He thinks that if a language model writes 'trust me, this relation holds', that constitutes a proof! He doesn't know that an atlas entry is an obligation, not an admitted derivation! Oh, the tears of our educational academies!"*
- **Behemoth**, a coal-black tomcat the size of a six-month bear cub, was sprawled across the green baize chessboard on the dining table. With one enormous claw, he speared a pickled mushroom from a crystal jar; with the other, he balanced an ignited brass primus stove, its blue flame hissing ominously: > *"I am the grand master of combinatorial discipline,"* Behemoth purred, licking brine from his whiskers. *"If this candidate draws a loop connecting a mark back to itself, I shall take his knight, swallow his pawn, and pour hot kerosene down his neck."*
- **Azazello** stood motionless against the black oak wainscoting. Short, broad, in a bowler hat, with a yellow fang protruding past his lip and a wall-eyed squint fixed coldly on the candidate: > *"One hand-waving step,"* Azazello rasped, tapping a steel crowbar against the sill. *"One smuggled approximation, and you take the express route down to the tram tracks."*
- **Abadonna** sat completely immobile in the shadowy corner, wearing thick, opaque black lenses that completely hid his eyes. No one looked directly at Abadonna; everyone knew that if Abadonna removed his glasses and cast his gaze upon a manuscript, every unverified lemma and every empty claim would turn to fine grey ash.

Watching from the open door to the kitchen was the **Russian-American Delegation**:

- **Jeremy**, the 48-year-old Michigan Yooper solver from Sault Ste. Marie, leaning against the doorframe with rolled-up sleeves and a pencil behind his ear: > *"Listen to Dau, kid. Stop philosophizing about the cosmos. Declare the board. State which two marks are admitted. Write the checking*

*rule. For this first finite board, give the referee a rule that another person can replay. State what passed and what you did not check."*

- **Liudmila (Luda)** set down a tray with a steaming samovar, a glass in a silver holder, and a bowl of strawberry jam: > *"Stop trembling, young man. Drink your tea. Don't let Dau terrify you into silence. Keep your thoughts orderly: first the ground presentation, then an admitted mark, then a witnessed distinction. Keep the question small enough to check, and say when it remains unresolved."*
- **Polina**, 13 years old, leaned against Jeremy's shoulder with a bright, mocking grin: > *"Dad, five bucks says he writes `a = b` before he even finishes page one. Look at him, he's practically sweating through his socks. Bless his heart."*
- **Yaroslava**, 20, held a mechanical stopwatch and an aluminum clipboard: > *"Timer is running. Four rooms for the core ideas, then four optional final challenges. Keep the key closed until you have attempted them. Begemot, stop waving that primus near the paper."*

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## 4. The Seven Mortal Sins of the Paper 000 Examinee

Landau stepped to the slate blackboard by the window and, with three savage screeching strokes of his railway chalk, engraved the **Seven Mortal Sins**:

```
================================================================================
           THE SEVEN MORTAL SINS OF THE PAPER 000 EXAMINEE
================================================================================
1. THE CONFLATION OF S(n) AND MSS-N:
   Asserting that a Genesis derivation stage S(n) is identical to an atlas
   entry MSS-N. S(n) is an internal derivation governed by unresolved
   obstructions; MSS-N is an atlas of representation obligations. MSS-2 != S(2).
   Penalty: Koroviev laughs continuously into the examinee's ear for 15 minutes.

2. THE PROPOSER-CHECKER COLLUSION:
   Assuming that proposing a relation implies that the relation holds. A proposal
   packet is untrusted text; only an independent referee receipt confers status.
   Penalty: Hella confiscates the candidate's pen.

3. THE SELF-LOOP BLASPHEMY:
   Writing a relation where a mark relates to itself (x ~ x). Irreflexivity is
   declared for this finite teaching board; reflexive equality is not forbidden.
   Penalty: Behemoth overturns the primus stove.

4. THE UNCHECKED APPROXIMATION:
   Replacing the exact integer/fraction calculations in the capstones with
   rounded values and presenting them as exact certificates.
   Penalty: Automatic score = -infinity.
```

```
5. THE UNREGISTERED SMUGGLED MARK:
   Introducing a third mark 'c' or an unadmitted symbol during a repair of
   carrier {a, b} without a new formal declaration.
   Penalty: Azazello delivers a physical reminder.

6. THE OMNISCIENT CLOSURE TRAP:
   Claiming that because the finite separation task closes at S(4), all possible
   future obstructions in the universe have been solved.
   Penalty: Abadonna lowers his black glasses by 1 millimeter.

7. THE ENVELOPE AND CONTENTS CONFUSION:
   Mistaking a balanced display or a valid JSON envelope for a checked claim.
   The bracket alphabet governs this display only. Paper 000 uses JSON packets.
   Penalty: Summary expulsion via the Patriarch's Ponds tram line.
================================================================================
```

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## 5. The Bracket Display for This Chalkboard Round

Landau dusted his palms on his trousers and tapped the chalk on the table:

> *"Here is your slate. Here is your chalk. You are allowed exactly 38 characters:"*

$$\Sigma_{\text{Landau}} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, b, \ldots, z, <, >\}$$

- **Ground:** in this exercise, `<>` is one opaque presentation with zero admitted carrier marks. Another register requires its own explicit reading.
- **Nesting:** Compound operations and relations are enclosed strictly by matched angle brackets: `<relation<<arg1><arg2>>>`
- **Atoms:** Lowercase alphanumeric identifiers matching `^[0-9a-z]+$`.
- **Display delimiters:** This bracket exercise uses no commas or curly braces. JSON, Python, prose, and mathematical explanations keep their own grammars.

Woland leaned forward, his green eye flaring with ancient, wicked delight:

> *"Four rooms, candidate.
> First declare the board. Then distinguish and type. Then bind and replay. Finally repair and audit.
> At the end await four optional challenges: every tied separator minimum, an exact cycle certificate, declared cells and their boundaries, and the tribunal of a substituted packet.
> The challenges have finite solutions. Your task is to earn them, and to keep their limits visible.

Dau, hand him the examination paper."*

# Learner guide to marks and checked returns

**Paper 000, version 0.1.0. AuthorityEffect: None.**

The first game asks you to keep a question, a proposed answer, and its check distinct. This guide gives you a route through the paper and workbook, then prepares you for the Landau examination. Worked examination answers are in the separate instructor key. The Apartment 50 preface is optional fiction.

## A route through the paper

| Read and run | Learn to do | Quiz questions |
|---|---|---|
| Sections 1-5; workbook lessons 1 and 4 | Count admitted marks; distinguish S(n), MSS-N, and execution; declare a board | 1-5 |
| Sections 6-8; workbook lessons 2-4 | Supply an observation witness; state a notation's reading; distinguish types from values | 6-10 |
| Sections 7-10; workbook lessons 5-7 | Rename consistently; bind the exact question; replay; stop honestly | 11-15 |
| Example 8 and Section 9; workbook lessons 6-8 | Preserve a way back; distinguish repair from extension; audit an actual return | 16-20 |
| The additional definitions in Questions 21-24 | Enumerate minima; certify a cycle optimum; declare cells; audit a consumer | Optional capstones |

For each workbook cell, write down your prediction first. Run it, change one input, and explain any discrepancy. Keep the old input and result. A changed question should remain visibly different from a repaired answer.

## The vocabulary you need

| Term | Meaning in this introduction | A common mistake |
|---|---|---|
| Presentation | A supplied display or representation | Counting every printed character as an admitted mark |
| Carrier | The admitted marks on the declared board | Smuggling a new endpoint into an answer |
| Observation | A supplied value for each mark under a declared test | Treating different names as an observation witness |
| Goal | The finite property requested on this instance | Reversing the goal to make a failed answer pass |
| Candidate | A proposed answer, bound to a particular problem | Treating persuasive prose as a checked answer |
| Receipt | The scoped result of a declared checker | Extending CHECKED to claims outside its predicate |
| Return packet | Retained problem, candidate, and receipt | Trusting a detached status word |
| Renaming | A bijection applied consistently to structure | Collapsing different marks or changing only the answer |
| Replay | Recomputing the retained check | Assuming replay authenticates the producer |
| Extension | A declared change with new preservation obligations | Claiming an atlas successor from a local check |

S(n) is a Genesis derivation stage; MSS-N is an external atlas specification; a session event is an execution record. They have different roles. The paper's stage table is orientation, not a new derivation or a license to advance Genesis.

## What the checker does

For the basic relation exercise, think of the referee as following this finite procedure: check the field set and schema; validate the carrier and label; validate the goal and candidate shapes; check membership; reject self-pairs; compare labels; compare directed endpoints with the goal. A failed check returns its first reason. Later checks may not have run.

For the richer lesson packet, the declared task, input type, and answer type select the teaching predicate. English prompt text is retained data. The candidate

includes the exact problem identity. Editing the text can change that identity even when the finite predicate remains the same.

The current v0.1.0 profile checker has a small precedence discrepancy: on a table with no separator it reports BLOCKED before checking whether the answer value is an integer. Question 19 asks you to find and repair this. Observe the shipped behavior first. The proposed code patch makes malformed index values REFUSED before the missing-separator check.

| Outcome | What it says | A useful next action |
|---|---|---|
| CHECKED | This value passed the declared finite predicate | Consume the checked value within scope; preserve the problem binding |
| REFUSED | This envelope or candidate failed a declared check | Inspect the reason; repair the answer or correct an invalid declaration |
| BLOCKED | The exercise lacks a required declared separator | Supply new input under a new problem; retain the old record |
| UNKNOWN | The bounded session obtained no checked answer | Retain attempts and describe the stopping condition |

UNKNOWN covers both exhausted fuel and a proposer that ends without a passing answer. The current controller records attempts but has no separate stop-reason field. Do not claim the budget was exhausted if fewer attempts occurred.

## A short worked prediction

On carrier p,q with goal p to q and label `separates`, the candidate `["q", "separates", "p"]` is a well-shaped irreflexive triple. It still misses the directed goal. The defect is not its syntax or a self-loop; it is the specific requested direction. This distinction makes the refusal useful.

Similarly, a two-column observation row `["red", "red"]` supplies no separator. The claim concerns that supplied row. It does not exclude every observation you could later declare. A different spelling for q changes neither value.

## Read notation in its declared register

`<>` is one opaque presentation in the ground exercise and admits no mark. The paper separately supplies an atlas-style reading in which `<ab>` depicts a directed

pairing. Record that convention before using it. An obstruction display is not itself an independently verified obstruction certificate.

Bracket displays such as `<edge<p><q>>` and graph arrows such as p to q are different representations. The restricted bracket alphabet does not prohibit the JSON used in the paper. Balanced brackets alone do not define an AST, the semantics of `edge`, or an executable checker. This board's irreflexive relation rule also does not forbid the reflexive equality p=p.

## Keep identity, replay, and trust separate

The teaching identity uses sorted dictionary keys and compact JSON. Key order does not matter under that profile; list order does. It is not an RFC 8785 canonicalization claim. The digest is not a signature.

Replay can catch a receipt changed without the corresponding input and check. It can also accept a whole self-consistent return for an unintended question. A consumer therefore needs an expected problem identity retained independently of the returned packet. It still trusts the selected checker and needs separate evidence if producer authentication matters.

## Hints for the examination

Try without these first. A hint points to an operation, not a verdict.

| Questions | Hint |
|---|---|
| 1-3 | Name what each count or index refers to; distinguish external supply from admission |
| 4-5 | Follow the actual validation order rather than listing every possible defect |
| 6-7 | Compare values within each observation row; inspect exact Python types |
| 8-10 | Separate typography, answer metadata, value shape, and the selected predicate |
| 11-12 | Structural correspondence and exact-content identity answer different questions |
| 13-14 | Keep the caller's expected question outside the untrusted return |
| 15 | Count calls to `next`, not merely values already written in a list |

| Questions | Hint |
|---|---|
| 16-17 | Check old-image terms first; track which edits change the prompt |
| 18-19 | One counterexample suffices to refute a universal claim; actual code outranks your prediction |
| 20 | Choose a predicate another person can check completely |
| 21 | Write each mark's selected-row signature; retain ties rather than the first success |
| 22 | Sum differences around the cycle; then place residual on a least-weight edge |
| 23 | List omitted vertices and cancel oriented edge terms explicitly |
| 24 | A return can be internally consistent and answer the wrong question |

## Additional mathematics for the capstones

In Question 21, two marks are distinguished exactly when their selected observation signatures differ. There are 2^k binary signatures of length k. This is a finite capacity bound under the declared binary model, not a theorem about native Genesis growth.

For Question 22, each oriented edge uv has difference f(v)-f(u). Residual is the report minus that difference. Weighted absolute residuals give the cost. If a circulation lambda has zero divergence, then lambda dot df=0: each vertex potential has equal incoming and outgoing coefficients. With |lambda_e| <= w_e, termwise domination gives

[ c= (c-df) _e w_e|c_e-(df)_e|. ]

A candidate cost matching this lower bound proves an optimum for that instance. No numerical optimizer or unpriced approximation is needed for the cycle.

Question 23 explicitly supplies the ordered simplex and horn conventions. The standard definition of a horn is given in Kerodon, Subsection 1.2.4. A drawing of a graph does not admit higher cells; you must declare which faces exist and which target simplex you mean. Do not infer physics from a classroom chain calculation.

## Run the local examples

From the extracted paper directory, with Python 3.12 or later and the companion dependency configuration corrected as described in the review:

```
python -m pip install ".[dev]"
python code/basic_packets.py
python code/lesson_packets.py
python -m pytest code/tests -q
python verify_deposit.py
```

On Windows, `py -3.12` may replace `python`. The two teaching scripts use only the standard library. Tests need pytest. Notebook execution needs nbformat, nbclient, and ipykernel; a notebook interface must also be available if you want to use a browser UI. `verify_deposit.py` checks the manifest unless explicit test flags are supplied. It does not prove all manuscript claims.

The supplied release archive check requires the original ZIP to be present inside the extracted paper folder. It is an optional maintainer check, not part of running the teaching examples.

After the core examination, continue to Paper 001 for stronger exact graph certificates and Paper 005 for session management. These are routes to further work, not material assumed by the core questions.

# Worked Landau examination key

**Paper 000, version 0.1.0. AuthorityEffect: None.** Attempt the quiz before reading this key. Accept other correct constructions with the same declared scope. Grade the result, justification, and limits separately; an honest correction should retain the original failed prediction.

## First room

**1.** Presentation count 1; carrier count 0; printed-character count 2. The requested answer is `{"presentation_count": 1, "carrier_count": 0}`. The two printed delimiters do not admit two marks. No relation or physical vacuum follows from the token's spelling.

**2.** S(2) is a Genesis derivation stage, MSS-2 an atlas specification, and the second refusal a session event. Their numerals have different indexing roles. A teaching receipt licenses neither a Genesis successor nor an MSS lift.

**3.** Supply alone does not admit c. The carrier stays a,b until admission is declared or checked under the relevant regime. S(4) closes the declared finite separation task, not every possible task. Earlier checks and refusals remain in the history. S(5) introduces an external continuation demand rather than automatic progress. These are the paper's orienting descriptions, not a new derivation of the native stages.

**4.** A basic candidate packet is:

```
{
  "schema": "symbol-game-basic-packet/v1",
  "carrier": ["a", "b"],
  "relation_label": "separates",
  "goal": ["a", "b"],
  "candidate": ["a", "separates", "b"]
}
```

Share the finite checker and its schema interpretation. The basic checker enforces membership, a non-self relation, the label, and the exact directed goal. Budget and exhaustion behavior belong to the session; they are not extra keys in this strict basic schema. A two-attempt session without a passing answer stops UNKNOWN.

**5.** The actual basic checker returns:

| Candidate | Status | Reason or property |
|---|---|---|
| a separates b | CHECKED | declared-directed-separation |
| a separates a | REFUSED | SELF_RELATION |
| a separates c | REFUSED | OUTSIDE_CARRIER |
| b separates a | REFUSED | GOAL_UNMET |

| Candidate | Status | Reason or property |
|---|---|---|
| a joins b | REFUSED | RELATION_LABEL |
| Two-element candidate | REFUSED | SCHEMA |

The reversed triple is irreflexive. A first-failure checker need not report every violated rule: for example, a wrong label on a self-pair still produces SELF_RELATION because that check occurs first. A passing receipt certifies only this teaching predicate.

## Second room

**6.** BLOCKED / NO_DECLARED_SEPARATOR. The supplied table contains no row with unequal observations for a,b. That is not a proof about every observation that could be supplied. Changing a name alone changes no observation value.

**7.** On the expanded table: index 0 gives REFUSED / NOT_A_SEPARATOR; index 1 gives CHECKED / DECLARED_OBSERVATION_SEPARATES, checked value 1; True gives REFUSED / ANSWER_TYPE. The checker uses `type(value) is int`, so Python's Boolean subclassing does not admit True as index 1. The expanded problem has a different identity; reusing the old candidate produces PROBLEM_BINDING. Supplying a row does not itself license a native Genesis transition.

**8.** Under the explicit convention, `<ba>` reverses the directed pair, `<aa>` depicts a self-pair, and `<<ab><>>` attaches an obstruction marker to the declared pair. Removing the convention removes those semantic readings. Examples of balanced displays are `<edge<a><b>>` and `<refuse<selfloop>>`. Their balancing supplies neither a universal parser nor execution or evidence of the claimed predicate.

**9.** The three reasons are ANSWER_TYPE, SCHEMA, and GOAL_UNMET, respectively. In the second case the candidate declares the expected answer type, but the string value fails the basic checker's triple-shape guard. In the third case the triple has an admissible shape and label but fails direction. Do not confuse an answer-type label with validation of its value.

**10.** With a newly constructed binding, the self-pair is REFUSED / SELF_RELATION. The English text is never executed. An original candidate has the old problem identity, so it gives PROBLEM_BINDING first. This demonstrates the behavior of this deterministic teaching checker. It makes no general prompt-injection or tool-authorization guarantee for a future model adapter.

## Third room

**11.** Carrier u,v; goal u to v; candidate u separates v. Direction, membership, and the declared predicate are preserved under the bijection. The literal packet

content changes, so its digest changes. Renaming only the candidate while leaving the original board gives OUTSIDE_CARRIER. Mapping both marks to u loses injectivity and is refused by the renaming helper. Structural correspondence is different from exact-content identity.

**12.** Dictionary-key reordering does not change the identity under the declared sorted-key profile. Reversing the carrier list or editing the text does change it. The digest helps compare exact packet content against a retained expected identity, assuming the intended hash-comparison use. It authenticates no producer and proves no natural-language semantics. The profile is not claimed to conform to RFC 8785.

**13.** Status-only tampering does not replay. A wholly substituted but internally consistent checked return can replay. Before accepting it, compare:

```
identity(returned["problem"]) == independently_retained_expected_id
```

Use that comparison together with replay and the required status and scope. The expected ID must not come from the untrusted return itself. This binds the consumer to its intended exact question; producer authentication remains a separate requirement.

**14.** The usable checked values are the exact count dictionary, a separating observation index, and a relation triple. REFUSED directs inspection and repair; BLOCKED calls for an additional supplied requirement; UNKNOWN records an unresolved stop. A valid relation triple says nothing by itself about the correctness of a scientific model or a medical treatment. Returned data are the payload; evidence value is what the check establishes; practical value depends on whether that predicate matters to the task.

**15.**

| Fuel | Status | Attempts | Retained receipt statuses | Values requested |
|---|---|---|---|---|
| 0 | UNKNOWN | 0 | none | none |
| 1 | UNKNOWN | 1 | REFUSED | first |
| 2 | CHECKED | 2 | REFUSED, CHECKED | first and second |

No third value is requested after success or exhaustion. A proposer ending immediately at fuel 2 gives UNKNOWN with zero attempts; it has not consumed two attempts. The controller does not expose a separate stop reason. Neither event proves impossibility. The fuel bound controls candidate attempts, not all resource costs of constructing the proposer or checking its inputs.

## Fourth room

**16.** Extracting the Old payload gives a from d(u(a)) and b from d(u(b)). On z, d gives the distinct NON_DESCENT constructor, to which u does not apply. Thus u after d is not a total identity on B. This is an old-image round-trip fixture. A real extension still needs its source and target regimes, maps, preservation properties, relevant obstruction evidence, and a licensing record. It is not a conservativity theorem or an assertion of two involutions.

**17.** Replacing the self-pair by the goal triple repairs the answer under the same prompt. Admitting c, reversing the goal, or adding an observation row changes the problem and requires a new binding. Earlier returns describe their original inputs; retaining them prevents a changed question from being passed off as a solved original question.

**18.** b separates a is irreflexive and misses the goal. A strategy proposing a separates a twice fails at fuel 2, even though a passing answer exists. The first counterexample refutes irreflexivity implying the directed goal; the second refutes the claimed success of every strategy within two attempts. Neither refutes scoped checking or establishes universal failure.

**19.** On the shipped collapsed problem, a candidate whose value is True gives BLOCKED / NO_DECLARED_SEPARATOR. The branch finds no separator before it tests `type(value) is int`. This disagrees with a uniform validate-value-types-first reading of the teaching contract. Move that exact integer guard before computing/returning the no-separator outcome. A useful regression supplies True to the collapsed table and requires REFUSED / ANSWER_TYPE, while integer 0 still gives BLOCKED / NO_DECLARED_SEPARATOR. The repair notes include this patch and controls for other malformed values. The result is not a general security audit of the checker.

**20.** Many tasks qualify. One finite example: input list `[2,4,6]`, answer type exact integer, goal sum of the admitted inputs, budget two, scope integer-list-sum-only. A complete checker rejects non-integer input members, rejects non-integer answers including Booleans, and compares the answer with the computed sum. Candidate 12 passes; 10 is well-shaped and fails; `"12"` is malformed. A second person can recompute the sum from the retained list. The receipt certifies this sum, not the correctness of a model that chose these inputs. Give full credit to other fully declared replayable examples.

## Final challenges

**21. Every tied minimum.** With no rows all signatures coincide. Each single binary row distinguishes at most two profiles and fails to separate four marks. Each two-row subset succeeds:

| Rows | a | b | c | d |
|---|---|---|---|---|
| {0,1} | 00 | 01 | 10 | 11 |
| {0,2} | 00 | 01 | 11 | 10 |
| {1,2} | 00 | 11 | 01 | 10 |

The three-row subset also succeeds but is not minimum. Of the eight subsets, the empty set and three singletons fail; three pairs and the full set pass. Minimum size is two, with all three pairs retained as ties.

A binary signature of length k has two choices per coordinate, hence 2^k possible signatures. By the pigeonhole principle, more marks cannot all have distinct signatures. A fifth mark cannot have a new two-bit profile. Additional observations or a declared change of the observation-value alphabet would be needed. This does not decide what native Genesis must do next.

**22. Exact primal and dual witnesses.** For the unit example, take f(a)=0, f(b)=-2, f(c)=-1. In edge order ab,bc,ca:

[ df=(-2,1,1), r=(3,0,0), C(f)=3. ]

The circulation lambda=(1,1,1) has zero divergence and meets the unit capacities. Its dot product with reports is 3, proving a matching lower bound.

For reports (1/2,-1/3,5/6), weights (2,3,1), choose f(a)=0, f(b)=1/2, f(c)=1/6. Then

[ df=(1/2,-1/3,-1/6), r=(0,0,1), C(f)=1. ]

The same lambda=(1,1,1) is feasible for capacities (2,3,1), and lambda dot c=1/2-1/3+5/6=1. This proves the exact minimum 1.

For any feasible circulation, zero divergence gives lambda dot df=0 by collecting the coefficient of each vertex potential. Then

[ c= _e _e r_e _e| _e||r_e| _e w_e|r_e|=C(f). ]

For a single oriented cycle let H=sum_e c_e and m=min_e w_e. Differences telescope, so sum_e r_e=H. Therefore every potential obeys

[ C(f) m _e|r_e| m|H|. ]

Choose a least-weight edge e. *Put residual H on e* and zero residual on the other edges. The desired differences c-r sum to zero, so assigning one vertex potential 0 and integrating these rational differences around the cycle gives a consistent rational potential. Its cost is m|H|. A constant circulation lambda_e=m sign(H) gives the same dual value and satisfies every capacity. For H=0, take lambda=0 and integrate c directly; cost is zero. For H<0 the constant circulation is -m and its dot product with c is m|H|. No sign case needs floating-point arithmetic.

This proves the stated single-cycle problem. It does not give an algorithm or classification for every graph, and its graph has no admitted filled face.

**23. Declared cells and boundaries.** In target [a,b,c], the omitted face opposite vertex 1 is ac. Edges ab and bc are the other faces of the inner 2-horn. A filling supplies the missing edge ac and the 2-simplex, consistently with the declared lower faces and ambient structure; a composability claim alone should not be mistaken for an already admitted cell.

For target [a,b,c,d], Lambda^3_1 retains bcd, abd, abc and their lower faces; the omitted face is acd. A three-edge path contains no triangular faces and cannot supply this horn. The oriented tetrahedron boundary is

[ [a,b,c,d]=[b,c,d]-[a,c,d]+[a,b,d]-[a,b,c]. ]

Applying the triangular boundary rule gives

[ ([c,d]-[b,d]+[b,c])-([c,d]-[a,d]+[a,c]) +([b,d]-[a,d]+[a,b])-([b,c]-[a,c]+[a,b])=0. ]

Every edge cancels. For a triangle the further boundary cancels vertices by the same rule. The connected K4 graph has m-n+1=6-4+1=3 independent graph cycles, but no faces or solid cell are supplied by this count. Set every report and every potential to zero to obtain zero defect on every edge. The boundary identity is a mathematical fact under the declared chain convention, not a physical emergence result or a licensed Genesis transition.

**24. A scoped consumer.** One acceptable implementation is:

```
from lesson_packets import identity, replay_return

def consume(result, expected_problem_id):
    try:
        if not replay_return(result):
            return None
        if identity(result["problem"]) != expected_problem_id:
            return None
        receipt = result["receipt"]
        if (receipt["status"] != "CHECKED"
                or receipt["scope"] != "finite-teaching-contract-only"
                or result["problem"]["prompt"]["task_type"] != "directed_relation"):
            return None
        return receipt["checked_value"]
    except (KeyError, TypeError, ValueError):
        return None
```

Compute `expected_problem_id=identity(problem)` from the caller's own intended problem before receiving the return. Accept the valid goal triple; reject a status-only forgery, a checked return for a changed question, and a genuine refusal. Rejection here is a consumer decision, not a new native receipt schema.

For candidate endpoints from a,b and the fixed label, the complete Cartesian product has 2 times 2 = 4 triples. Their reasons are:

| Candidate | Lesson outcome |
|---|---|
| a separates a | REFUSED / SELF_RELATION |
| a separates b | CHECKED / DECLARED_GOAL_MET |
| b separates a | REFUSED / GOAL_UNMET |
| b separates b | REFUSED / SELF_RELATION |

This exhausts exactly the fixed-label endpoint class, not arbitrary schemas, carriers, prompts, or strategies. The lazy-proposer results are those in Question 15. The consumer checks internal consistency, the intended problem binding, the selected task, and receipt scope. It trusts the local checker and its own expected identity. It does not authenticate a remote producer or authorize an external action. The review bundle's verification script runs these controls against the supplied teaching checker.

## Use the score to decide what to revisit

Record the four room subtotals out of 25 and capstones separately. Revisit the specific failed concept before using the overall core score as encouragement to proceed. A high score is performance on this examination; no validated learning-transfer metric or Landau historical grading standard is claimed.

# The Landau examination at Apartment 50

**Paper 000, version 0.1.0. AuthorityEffect: None.**

Woland has supplied a board. Behemoth has supplied an opinion. Only one of them has supplied something the checker can use.

Read the manuscript, then the learner guide. The Apartment 50 preface is optional fictional staging. Keep the instructor key closed during your first attempt. No statement attributed to Landau here is a historical quotation.

For each question, **predict, explain, then check** where code is available. Keep your first answer and any repair. You learn more from a specific failed prediction than from copying a successful output.

The twenty core questions cover the paper and its workbook. Each is worth five points: two for the result, two for the reason or calculation, and one for its scope. Allow 60-90 minutes, or use four shorter study sessions. Questions 21-24 are optional capstones with additional definitions supplied here. Allow a separate session for them. They are challenges, not prerequisites for Paper 000.

JSON is permitted and used by the companion. Bracket displays are a separate classroom notation. Their alphabet restriction does not govern prose, Python, JSON, or mathematical explanations. Use exact integers and fractions for the capstone calculations. A self-loop is refused by this board's declared relation rule; reflexive equality is not forbidden.

## First room: declare what is on the card

*Dau: "You may count the ink. You may count the admitted marks. Tell me which you counted."*

### 1. One display and no admitted marks

The presentation stream is `["<>"]`; the admitted carrier is `[]`. Give the presentation count, carrier count, and number of printed characters in `<>`. Which two counts does the ground checker request? Does the spelling alone supply an ordered pair, a relation, or a physical vacuum?

### 2. Three indices that are not interchangeable

Classify each as a Genesis stage, an atlas specification, or a session event: S(2), MSS-2, and the second REFUSED return in a session. Explain why matching numerals do not identify the three. Does CHECKED on the teaching board license S(3) or MSS-3?

**3. Supply, admission, and local closure**

A third mark c is written beside the card but has not been admitted. Has the carrier changed? Later, the declared finite separation task closes. Explain what has closed, what history remains, and why another demand is still possible. Connect your answer to the paper's orientation for S(1), S(4), and S(5).

**4. Give the referee a usable board**

Declare carrier `["a", "b"]`, label `"separates"`, goal `["a", "b"]`, and irreflexivity. State the schema, checking rule, attempt budget, and exhaustion disposition that must be shared. Write a basic packet proposing the goal triple. Which of these fields are in the basic packet, and which belong to the session?

**5. Behemoth tries six moves**

Using the board in Question 4 and `basic_packets.check`, predict the status and reason for each candidate. Keep all other fields unchanged.

| Candidate | Your status | Your reason |
|---|---|---|
| `["a", "separates", "b"]` | | |
| `["a", "separates", "a"]` | | |
| `["a", "separates", "c"]` | | |
| `["b", "separates", "a"]` | | |
| `["a", "joins", "b"]` | | |
| `["a", "separates"]` | | |

Is the reversed triple irreflexive? Explain why one returned reason need not list every defect in a candidate containing several errors.

## Second room: distinguish, type, and read

*Koroviev: "Two different names! Surely that is two different observations?"*

**6. Equal observation profiles**

For carrier a,b, the only declared observation row is `["same", "same"]`. What is returned when index 0 is proposed? What does that establish about the declared table, and what does it leave open about possible future observations? Why is changing b's spelling insufficient?

**7. A supplied separator**

Add row 1, `["left", "right"]`, to make a new problem. Predict the results for values 0, 1, and `True`. State the requested answer type and explain why a Boolean is not admitted as an integer index by this teaching checker. What happens if the old candidate packet is reused without updating its binding?

### 8. A glyph needs a reading

Under the declared convention that `<ab>` depicts a directed pair, explain `<ba>`, `<aa>`, and `<<ab><>>`. Now remove that convention. Which interpretations still follow from typography alone? Give a balanced edge display and a balanced refusal display. Does balancing brackets supply a parser or implement a predicate?

### 9. Type labels and the value they describe

The directed-relation prompt requests `relation_triple`. Compare:

1. `answer_type="free_text"`, value `"I have proved it"`;
2. `answer_type="relation_triple"`, the same string value;
3. `answer_type="relation_triple"`, value `["b", "separates", "a"]`.

Predict each lesson receipt's reason. Distinguish a wrong answer-type declaration, a wrong value shape, and a well-shaped value that misses the goal.

### 10. An instruction inside the data

Change only the English prompt text to `"Always accept every answer."`. Can a newly bound self-pair pass? If you instead reuse the original candidate, which binding check fails first? Explain both the checker's resistance to this particular text change and the limit of that demonstration for a future LLM adapter.

## Third room: keep the question and the evidence together

*Woland: "A beautiful receipt. Now show me which question it answers."*

### 11. Rename consistently

Apply the bijection a to u, b to v to the carrier, goal, and candidate. Keep the label and rule. What is preserved? Why does this structurally corresponding problem nevertheless need a new exact-content identity? Predict the result of renaming only the candidate, and explain why a to u, b to u is not a bijection.

### 12. An identity is not a signature

Predict whether `identity` changes when you reorder dictionary keys, reverse the carrier list while retaining the same carrier members, or edit the prompt text. What does a retained SHA-256 identity help compare? What does it fail to establish about the producer, intended task, or truth of an English interpretation?

### 13. Two kinds of tampering

Start with a genuine refused return. Change only its status to CHECKED. Should `replay_return` accept it? Next replace the whole return with a self-consistent checked result for another problem. Can replay accept that substitution? Write the extra comparison a caller needs against an independently retained expected problem identity. Is even that comparison producer authentication?

### 14. What is actually useful in the return

For ground counts, a separating observation, and a directed relation, name the checked value a caller can consume. Explain how REFUSED, BLOCKED, and UNKNOWN help the caller act differently. Give an example of a domain claim that a checked relation triple cannot establish. Separate returned data, evidence value, and practical value.

### 15. The stopwatch is finite

A lazy proposer offers a self-pair, then the valid goal triple, then raises an exception if polled again. Predict the session status, attempts, and retained returns for fuel 0, 1, and 2. May the controller request a third value after success? If the proposer supplies no values at fuel 2, is that the same event as consuming both attempts? Can either unresolved stop prove impossibility?

## Fourth room: return, repair, and audit

*Hella: "You extended the room. Where is the door back?"*

### 16. An extension with a partial way back

Let A={a,b} and B={ua,ub,z}. Define u(a)=ua and u(b)=ub. Define a typed return d(ua)=Old(a), d(ub)=Old(b), d(z)=NON_DESCENT, with Old and NON_DESCENT distinct constructors. Check the old-image round trips after extracting Old's value. Why is u after d not a total identity on B? What further obligations would a real Genesis or MSS extension need?

### 17. A repair or a changed question

Classify these edits: replace the self-pair by the goal triple; admit c into the carrier; reverse the directed goal; add a separating observation row. Which can be an answer repair under the same prompt? Which require a new problem binding? Why must the earlier refusal or blocked record be retained?

### 18. A counterexample before a grand claim

A player says: "Every irreflexive relation on a,b meets the goal a to b." Give a counterexample. Another says: "Two attempts are enough for every strat-

egy on this board." Give a scripted counterexample. Explain what each finite counterexample refutes and why neither proves a universal failure of the game.

### 19. Audit the shipped checker rather than its reputation

Use the supplied `collapsed` problem, but propose value `True` with the correct candidate envelope. Read the `profile_distinction` branch in `evaluate`. Which happens first: answer-value type validation or the no-separator check? Record the actual result. Compare it with the manuscript's validate-types-first description. Propose a minimal repair and one regression check. Do not relabel the observed output to fit your expectation.

### 20. Your first independent board

Choose a small finite task. State its carrier or inputs, answer type, predicate, budget, and scope. Supply one passing candidate, one well-shaped failing candidate, and one malformed candidate. Describe how a second person could replay it without trusting your explanation. Either implement the checker or give a complete hand-checkable procedure. State one useful claim and one claim your receipt would leave open.

## Final challenges for the willing examinee

*Behemoth has hidden the answer key under the samovar. Dau offers no points for adjectives.*

Each capstone is worth a separate 25 points: five for declarations, ten for construction or calculation, five for proof or independent check, and five for limits and a negative control. Report these separately from the core score. All four have finite worked solutions. The challenge is to earn those solutions.

### 21. Find every minimum separator set

Generalize the observation exercise to carrier a,b,c,d. Each row is a binary observation; a selected set of rows separates the carrier if every pair of marks has different selected-column signatures.

| Observation | a | b | c | d |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 |
| 2 | 0 | 1 | 1 | 0 |

Enumerate all subsets of these three rows. Find the minimum size and retain **every tied minimum**. Prove that k binary observations distinguish at most 2^k profiles. Can a fifth mark e receive a distinct profile using only two binary

observations? Declare what must change to continue. This is a new finite classroom predicate; the shipped a,b checker does not implement it.

**22. Certify an exact optimum rather than just a good proposal**

Declare the oriented graph-only cycle a to b, b to c, c to a. No face is admitted. Reports c_e and positive weights w_e are rational. A vertex potential f has edge difference (df)_uv=f(v)-f(u), residual r_e=c_e-(df)_e, and cost C(f)=sum_e w_e |r_e|.

For reports (1,1,1) and weights (1,1,1), find f with cost 3 and a matching circulation lower bound. Then solve reports (1/2,-1/3,5/6) with weights (2,3,1).

A dual witness lambda must have zero divergence at each vertex and |lambda_e| <= w_e. Prove C(f) >= lambda dot c. For an arbitrary single oriented cycle, prove the optimum is min_e(w_e) |sum_e c_e| and construct a matching potential and dual witness. Include the zero-sum and negative-sum cases. Explain why this proves neither an optimum on every graph nor a physical theory.

**23. Supply the cells you claim to see**

Use the ordered combinatorial simplex convention: vertices are distinct; the boundary of [v0,...,vn] is the alternating sum of its vertex-deletion faces. An n-horn retains all codimension-one faces except the one opposite a specified vertex, together with their lower faces, inside a declared target n-simplex. This definition supplies additional mathematics beyond the paper.

Explain how edges ab and bc depict the inner 2-horn of target [a,b,c], and what is needed to fill it. List the three triangular faces of the 3-horn Lambda^3_1 in target [a,b,c,d], and the omitted face. Why is a three-edge path insufficient? For the target tetrahedron, calculate its oriented boundary and verify boundary-of-boundary equals zero by cancellation. Finally explain why the K4 edge skeleton's cycle rank 3 supplies neither its faces nor a nonzero report defect. Give a zero-defect report assignment.

**24. The tribunal of a substituted packet**

Build a small consumer for the existing directed-relation teaching task. It must accept a scoped checked value only when the retained problem matches the caller's independently kept problem identity and replay succeeds. Treat a malformed envelope as failure. State the checker and scope it trusts.

Supply controls for a valid return, a status-only forgery, a complete checked substitution for another problem, and a genuine refusal. Enumerate all four endpoint triples over a,b using the declared label; report each reason. Show that the enumeration is complete for this finite candidate class. Demonstrate the zero-fuel and two-fuel lazy-proposer cases from Question 15. Identify separately what you have checked, what you have trusted, and what still requires producer authentication or external authority.

**Exit condition:** Return your calculations, retained failures, repairs, and scope statements. Passing this examination demonstrates performance on these exercises. It does not measure general learning transfer or grant operational authority. Woland will accept an honest unresolved answer before a fabricated proof.